

Solution of Exercise Sheet 6

Exercise 1 (File Systems)

1. Describe the information inodes store.

An inode stores a file's metadata, except the file name.

2. Name three examples of metadata in the file system.

Metadata are among others the size, UID/GID, permissions and date.

3. Describe what a cluster in the file system is.

File systems address clusters and not blocks of the storage device. Each file occupies an integer number of clusters.

4. Describe how a UNIX file system (e.g. ext2/3), which does not implement extents, can address more than 12 clusters.

If a file requires more than 12 clusters, these clusters are indirectly addressed via clusters that contain only cluster numbers.

5. Describe how directories in the Linux file systems are technically implemented.

Directories are just text files, which contain the names and inodes of files.

6. Explain why moving a large file within a file system is always faster than copying it.

Directories are just text files containing the names and inode numbers of files. Moving a file within a file system implies removing a line in the old directory and adding a line in the new directory. On the other hand, copying a file creates a new inode and duplicates the file contents.

7. Name one advantage and one drawback of small clusters in the file system compared with large clusters.

Advantage: Decreasing capacity loss due to internal fragmentation.

Drawback: Rising overhead for large files.

8. Do DOS/Windows file systems differentiate between uppercase and lowercase?

☐ Yes ☒ No

9. Do UNIX file systems differentiate between uppercase and lowercase?

☒ Yes ☐ No

10. Do modern operating systems accelerate requests to stored data with a cache in the main memory.

☒ Yes ☐ No

11. Most operating systems operate according to the principle...

☒ write-back ☐ write-through

12. Name one advantage and one disadvantage of a cache in the main memory, which is used by the operating system to accelerate the requests to stored data.

Benefit: Better system performance.

Drawback: System crashes may cause inconsistencies.

13. Explain what an absolute path name is.

It is a path name, which describes the complete path from the root to the file.

14. Explain what a relative path name is.

It is a path name, which does not begin with the root.

15. `/var/log/messages` is an/a...

☒ absolute path name ☐ relative path name

16. `BTS_Vorlesung/Vorlesung_05/bts_slides_05_en.tex` is an/a...

☐ absolute path name ☒ relative path name

17. `Documents/MasterThesis/thesis.tex` is an/a...

☐ absolute path name ☒ relative path name

18. `/home/<username>/Mail/inbox/` is an/a...

☒ absolute path name ☐ relative path name

19. Describe what information the boot sector (also called boot block) of a file system stores.

The boot sector (boot block) contains executable machine code („boot loader“), which starts the operating system, and in some file systems like FAT12/16/32 some information about the file system too.

20. Describe what information the super block of a file system stores.

It contains information about the file system, e.g. number of inodes and clusters.

21. Explain why some file systems (e.g. ext2/3) do combine the clusters of the file system to block groups.

Inodes (metadata) are physically located close to the clusters, they address.

22. Describe what the File Allocation Table (FAT) is and the information it stores.

The FAT (File Allocation Table) is a table of fixed size. For each cluster in the file system, an entry exists in the FAT with the following information about the cluster:

- Cluster is free or the storage medium is damaged at this point.*
- Cluster is occupied by a file and stores the address of the next cluster, which belongs to the file or it is the last cluster of the file.*

23. Describe the objective of the journal in a journaling file system.

In the journal, write operations are collected before being committed to the file system.

24. Describe a benefit of using a journaling file system compared with using a file system without a journal.

After a crash, only the files (clusters) and metadata must be checked, for which a record exists in the journal.

25. Name the three values that are required to store an extent.

Start (cluster number) of the area (extent) in the file.

Size of the area in the file (in clusters).

Number of the first cluster on the storage device.

26. Describe the benefit of using extents compared with direct addressing of the clusters.

Instead of multiple individual clusters numbers, only 3 values are required: Lesser overhead.

27. Name one Linux file system that implements extents.

JFS, XFS, btrfs, ext4

28. Name one Windows file system that implements extents.

NTFS

29. When a file within a copy-on-write (COW) file system gets modified, the old clusters in the file system that need to be modified are...

- ☐ are overwritten with the new modifications.
- ☒ are preserved (not changed).
- ☒ are copied into new clusters, where the modifications are made.
- ☐ are erased, by removing the cluster address in the inode.

(Hint: Two answers are correct!)

30. Describe the result of defragmenting a file system.

Logically related clusters are placed physically close to each other on the storage device.

31. Describe the sort of data processing that is maximum accelerated by defragmenting.

A continuous arrangement would maximum accelerate continuous forward reading of data because no more seek times occur (when using Hard Disk Drives).

32. Describe the scenario where defragmenting is useful.

Only if the seek times are huge, defragmentation makes sense.

Exercise 2 (File Systems)

Please mark for each statement about file systems, whether the statement is true or false.

Statement	true	false
Inodes store all metadata of files.		X
File systems address clusters and not blocks of the storage medium or storage drive.	X	
The smaller the clusters are, the more overhead for large files occur.	X	
The bigger the clusters are, the lesser capacity is lost due to internal fragmentation.		X
In UNIX, file extensions have always been of great significance.		X
Modern file systems operate so much efficient that buffering by the operating system is no longer common.		X
Absolute path names describe the complete path from the root to the file.	X	
The separator in path names is identical for all operating systems.		X
An advantage of block groups is that the inodes are physically located close to the clusters, they address.	X	
For each cluster in the file system, an entry exists in the FAT.	X	
Because of the Master File Table in NTFS, fragmentation cannot occur.		X
The journal of journaling file systems reduces the number of write operations.		X
Journaling file systems narrow down the data, which need to be checked during the consistency check.	X	
When using journaling file systems, a loss of data is impossible.		X
If metadata and file contents are journaled both, all write operations are carried out twice.	X	
Extents cause lesser overhead compared with block addressing.	X	

Exercise 3 (Pattern Comparison and Data Analysis)

1. Name (or describe) one useful application for the command `sed`.

This tool is used for the parsing and transforming of text.

2. Create a file `sedtest.txt` with the following content:

Line 1
Line 2
Line 3
Line 4
Line 5
Line 6

```
$ echo -e "Line 1\nLine 2\nLine 3\nLine 4\nLine 5\nLine 6"  
> ~/sedtest.txt
```

Insert with `sed` 3 blanks at the beginning of each line.

```
$ sed "s/^/   /" ~/sedtest.txt
```

3. Print out with `sed` the lines 2 to 5 of the file `sedtest.txt`.

```
$ sed -n "2,5p" ~/sedtest.txt
```

4. Remove with `sed` each second line of the file `sedtest.txt`.

```
$ sed "n;d;" ~/sedtest.txt
```

5. Create a file `htmlcode.txt` with the following content:

```
<a href="BTSWS2019/index.html">Operating Systems (OpSys)</a><p>
<b>This is a <i>HTML file</i></b><br>
<h2>This is a headline</h2>
```

```
$ echo -e "<a href='BTSWS2019/index.html'>Operating Systems (OpSys)</a><p>" >> ~/htmlcode.html
$ echo -e "<b>This is a <i>HTML file</i></b><br>" >> ~/htmlcode.html
$ echo -e "<h2>This is a headline</h2>" >> ~/htmlcode.html
```

Remove with `sed` all HTML tags from the file `htmlcode.html`.

```
$ sed -e "s/<[^>]*>//g" ~/htmlcode.html
```

6. Create a file `umlaute.txt` with the following content:

```
Bäume, Äpfel, Bücher, Übertreibung
Töpfe, Öffentlichkeit, Straße, Spaß
```

```
$ echo "Bäume, Äpfel, Bücher, Übertreibung" >> ~/umlaute.txt
$ echo "Töpfe, Öffentlichkeit, Straße, Spaß" >> ~/umlaute.txt
```

Modify with `sed` all umlauts in the file `umlaute.txt` into „ae“, „oe“, „ue“, „Ae“, „Oe“, „Ue“ and „ss“.

```
$ sed -e "s/ä/ae/g;s/Ä/Ae/g;s/ö/oe/g;s/Ö/Oe/g;s/ü/ue/g;s/Ü/Ue/g;s/ß/ss/g" ~/umlaute.txt
```

7. Create a file `bundesliga_08_0405.txt` with the results of the 8th match day of the season 2004/2005:

Schalke	- Bochum	3 : 2	61500 spectators
Bielefeld	- Stuttgart	0 : 2	22700 spectators
Dortmund	- Nürnberg	2 : 2	73500 spectators
Leverkusen	- Hamburg	3 : 0	22500 spectators
Freiburg	- Mainz	1 : 2	24000 spectators
Kaiserslautern	- Berlin	0 : 2	30500 spectators
Wolfsburg	- Mönchengladbach	2 : 1	26500 spectators
Rostock	- Hannover	1 : 3	16500 spectators
Bremen	- München	1 : 2	42000 spectators

```
$ echo -e "Schalke      - Bochum      3 : 2 61500 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Bielefeld    - Stuttgart   0 : 2 22700 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Dortmund     - Nürnberg   2 : 2 73500 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Leverkusen    - Hamburg    3 : 0 22500 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Freiburg      - Mainz      1 : 2 24000 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Kaiserslautern - Berlin     0 : 2 30500 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Wolfsburg     - Mönchengladbach 2 : 1 26500 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Rostock       - Hannover    1 : 3 16500 spectators" >> ~/bundesliga_08_0405.txt
$ echo -e "Bremen       - München     1 : 2 42000 spectators" >> ~/bundesliga_08_0405.txt
```

8. Name (or describe) one useful application for the command `awk`.

This tool is used for text processing and data extraction and reporting.

9. Determine with `awk` all matches, which had more than 35000 spectators.

```
$ awk '$7 > 35000' ~/bundesliga_08_0405.txt
Schalke - Bochum 3 : 2 61500 spectators
Dortmund - Nürnberg 2 : 2 73500 spectators
Bremen - München 1 : 2 42000 spectators
```

10. Determine with `awk` all matches, which had less than 50000 spectators and where the home team won.

```
$ awk '$4 > $6 && $7 < 50000' ~/bundesliga_08_0405.txt
Leverkusen - Hamburg 3 : 0 22500 spectators
Wolfsburg - Mönchengladbach 2 : 1 26500 spectators
```

11. Determine with `awk` for each game the sum of the scored goals.

```
$ awk '{print $4+$6 " " " $1 $2 $3}' ~/bundesliga_08_0405.txt
5 Schalke-Bochum
2 Bielefeld-Stuttgart
4 Dortmund-Nürnberg
3 Leverkusen-Hamburg
3 Freiburg-Mainz
2 Kaiserslautern-Berlin
3 Wolfsburg-Mönchengladbach
4 Rostock-Hannover
3 Bremen-München
```

12. Determine with `awk` in which city the most spectators visited the match and print out the result this way:

The most spectators were in CITY (NUMBER).

```
$ awk -v max=0 'max<$7 {max=$7; city=$1} \
END{print"The most spectators were in",city, "("max")"}' \
~/bundesliga_08_0405.txt
The most spectators were in Dortmund (73500)
```